

DIGITAL FORENSIC INVESTIGATION REPORT

Deleted Data Recovery Examination

CASE IDENTIFIER	AIT-DF-2026-009
INVESTIGATION TYPE	Digital Forensics / Deleted Data Recovery
EVIDENCE TYPE	Forensic Disk Image
INVESTIGATOR	_____
TEAM / UNIT	_____
REPORT DATE	27 September 2026

Report status Working report structure based on the supplied examination record. Any item that was not independently reproduced should be verified against the actual evidence before final submission.

1. Executive Summary

This report documents a digital-forensic examination of a supplied forensic disk image associated with a suspected data-deletion incident. The examination focused on six target directories: SSH, My pic, Our pic, ADDS Project, Corel, and Corel Draw Work.

The supplied examination record identifies the image as an NTFS volume and records verification of the source image using SHA-256 and MD5. Analysis with FTK Imager and Autopsy identified deleted filesystem records and unallocated-space artifacts. The record states that Autopsy parsed 1,162 deleted filesystem entries and that several target graphic assets were recoverable.

Three target directories—SSH, Corel, and corel draw work—were reported as deleted directory entries visible from the root MFT structure. My pic, Our pic, and ADDS Project were reported as deleted and unallocated, meaning that their directory structure was not directly available in the active root directory and recovery depended on carving and related artifacts.

The strongest deletion-related boundary in the supplied record is 16 September 2026 at 11:01:30 WAT, when the relevant deleted records show an NTFS change timestamp. A later Recycle Bin artifact was recorded on 24 September 2026 at 11:42:43 WAT. These timestamps establish a defensible activity window, but they should not be presented as proof of the physical keyboard action or the identity and intent of the person who performed it.

Key conclusion The evidence supports prior existence, subsequent deletion, and varying degrees of recoverability. It does not, by itself, establish the human actor or subjective intent behind the deletion.

2. Case Background

Apex Integrated Technologies Ltd. reported that several folders previously associated with a workstation were no longer visible during routine operations. The investigation was commissioned to determine what could be established from the supplied forensic image without relying on assumptions about user behaviour.

3. Investigation Objectives

- Determine the status of the six target folders: present, deleted, partially deleted, recoverable, partially recoverable, or unrecoverable.
- Verify the integrity of the supplied forensic image and document the evidence-handling process.
- Identify relevant filesystem, deleted-file, metadata, and timeline artifacts.
- Recover relevant deleted material where the filesystem evidence permits recovery.
- Correlate observations across dd, R-Drive Image, FTK Imager, and Autopsy.
- Establish the most defensible timeframe for the deletion-related activity and state the limitations of that conclusion.

4. Scope

In scope	Out of scope
Forensic image verification and filesystem examination	Physical inspection of the original workstation
NTFS/MFT, deleted entries, unallocated space and Recycle Bin artifacts	Live network traffic capture
Target-folder and deleted-file analysis	Attribution of subjective intent
Metadata and timeline correlation	Claims unsupported by artifacts

5. Evidence Identification

Field	Recorded value
Evidence filename	Evidence.E01
Format	Expert Witness Format / EnCase image (.E01)
Filesystem	NTFS (New Volume)
Acquisition date	24 September 2026
SHA-256	6216149FDDF6EB7074349AFAEF226F02C3B246EE6C3C3AE91C5AB7ADEB8C29E0
MD5	ae39e9a30ab5a65809eb93289938e149

6. Integrity Verification

The supplied record states that SHA-256 and MD5 were calculated for the evidence and that a working copy produced matching values. The significance of this comparison is that the examined copy can be shown to correspond mathematically to the source value recorded at intake.

Question	Result recorded in supplied examination
Hash algorithms	SHA-256 and MD5
Working copy created?	Yes
Hash comparison	Matching
Unexpected hash change would indicate	Possible corruption, alteration, or handling problem requiring investigation

7. Chain of Custody

Field	Recorded value
Case ID	AIT-DF-2026-009
Evidence ID	EV-01
Evidence description	Forensic Disk Image (Evidence.E01)
Received from	Apex Integrated Technologies Ltd.
Received by	Ogooluwa Afolayan James
Date/time received	24 September 2026 @ 09:00 WAT
Storage location	Digital Forensics Secure Storage

8. Tools and Examination Method

Tool	Role in the examination
dd	Bit-for-bit copying / hash verification activity
R-Drive Image	Image and storage-structure verification
FTK Imager v8.3.0.27	Filesystem tree, MFT/deleted-entry, Recycle Bin and unallocated-space inspection
Autopsy v4.23.1	Deleted-file parsing, metadata review, carving/recovery and timeline analysis

9. Examination Workflow

1. Identify the supplied image and record its format, filesystem and cryptographic hashes.
2. Verify that the working copy matches the recorded source hash values.
3. Inspect the NTFS directory tree and identify the six target names.
4. Determine whether each target appears as an active entry, deleted MFT record, or unallocated/carved artifact.
5. Inspect \$RECYCLE.BIN for supporting deletion artifacts.
6. Parse deleted filesystem records in Autopsy and compare metadata across entries.
7. Attempt recovery of relevant deleted files and record their recovery status.
8. Correlate timestamps into a sequence rather than treating one timestamp as automatically equal to the deletion action.
9. Compare findings across tools and document limitations.

Forensic writing rule An observation describes what the software displayed. An interpretation explains what that observation is consistent with. A claim about human intent requires additional evidence.

10. FTK Imager Findings

The supplied FTK examination shows three target directories—Corel, corel draw work, and SSH—beneath the root of the NTFS volume with deletion indicators. The same examination also identified unallocated space blocks and a user-specific \$RECYCLE.BIN directory.

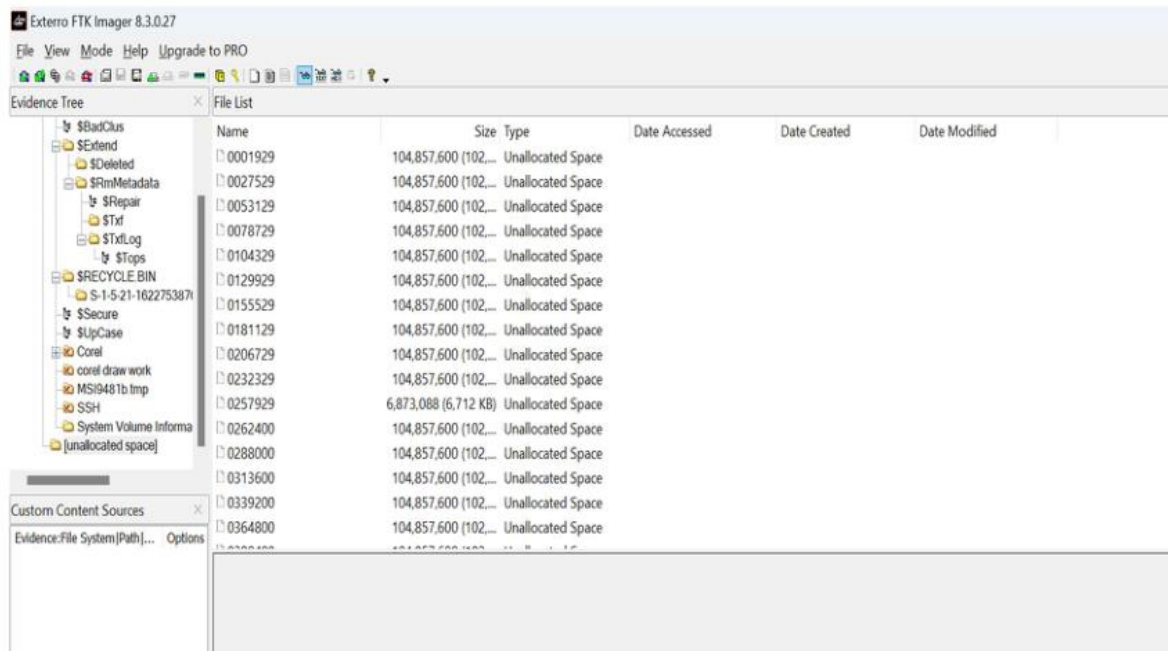


Figure 10.1 — FTK Imager view of unallocated-space blocks and filesystem tree. This is a reference exhibit carried over from the supplied examination record; replace with an independently captured screenshot if available.

The screenshot is useful for demonstrating where the target directories and unallocated-space branch were observed. It should be interpreted together with the underlying filesystem findings rather than as stand-alone proof of how the deletion occurred.

10.1 Recycle Bin Examination

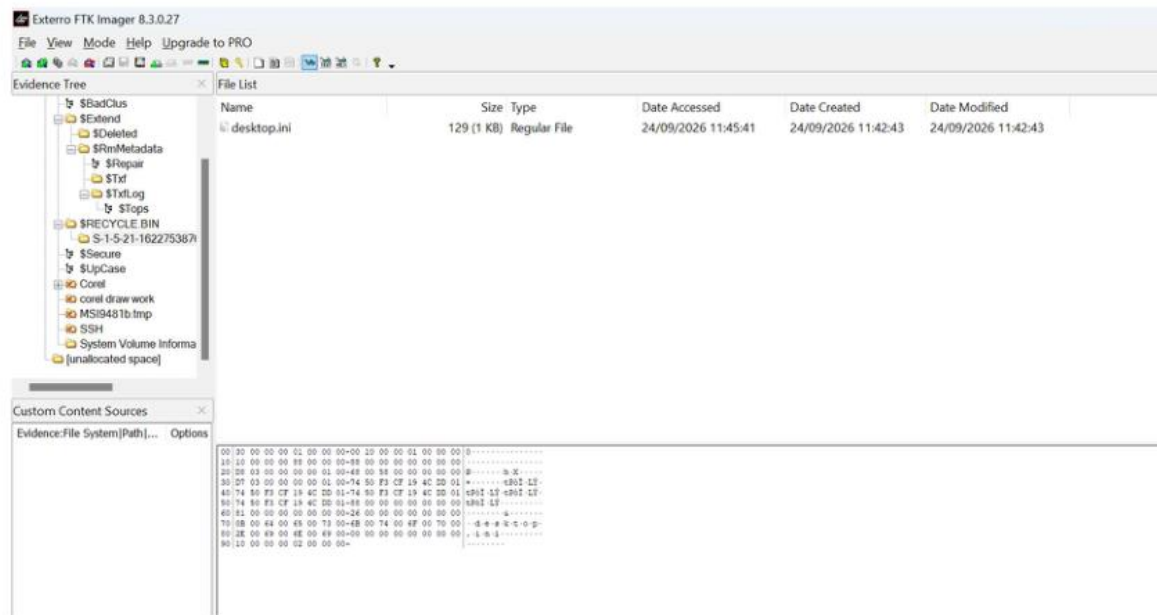


Figure 10.2 — FTK Imager view of the user SID Recycle Bin directory showing desktop.ini. Reference exhibit from the supplied examination record.

The supplied record reports that the relevant SID directory contained desktop.ini and no \$I/\$R records for the target folders. That absence is consistent with deletion that bypassed the normal Recycle Bin metadata path or with later purging, but the artifact alone cannot distinguish those possibilities.

11. Target Folder Status

Target	Filesystem status	Reported contents / artifacts	Recovery
SSH	Deleted MFT entry	SSH configuration material, keys and known-hosts records	Fully recoverable in supplied record
My pic	Deleted / unallocated	JPG/PNG photographic material	Recoverable through carving
Our pic	Deleted / unallocated	JPG/PNG photographic material and related artifacts	Recoverable through carving
ADDs Project	Deleted / unallocated	PowerShell scripts and project documentation	Recoverable through carving / keyword search
Corel	Deleted MFT entry	Corel PHOTO-PAINT .cpt assets	Fully recoverable in supplied record
Corel Draw Work	Deleted MFT entry	.cdr design material, exports and project drafts	Fully recoverable in supplied record

11.1 SSH

The supplied examination identifies SSH as a deleted directory entry in the root MFT structure. The associated material is described as configuration files, public/private keys and known-hosts records. The record states that the content was recoverable through FTK Imager export and Autopsy filesystem parsing.

11.2 My pic and Our pic

Both picture directories were reported as deleted and unallocated rather than remaining as visible root directory entries. The examination identified JPG/PNG material and reported recovery through Autopsy file carving. Because directory structure was not preserved in the same way as the root-level deleted entries, recovered files should not automatically be treated as proof of their original folder path.

11.3 ADDs Project

ADDs Project was reported as deleted and unallocated. The supplied record associates the recovered material with Active Directory project files, PowerShell scripts and project documentation. Carving and keyword search were used as the stated recovery approaches.

12. Corel-Related Findings

The Corel directory was reported as a deleted MFT entry. Named graphic assets include Buttons.cpt, Cereal.cpt, Cookies.cpt, DESIGN28.cpt, DESIGN29.cpt, Dice.cpt, fire.cpt and Flowers.cpt. The supplied metadata associates these records with a change time of 16 September 2026 at 11:01:30 WAT and an access time of 24 September 2026 at 12:40:08 WAT.

Corel Draw Work was likewise identified as a deleted directory entry, with .cdr files, exported graphics and project drafts reported as associated content. The supplied examination states that this material was recoverable.

13. Deleted Evidence Processing

Item	Original path / context	Status	Recovery method
Corel	\Corel	Deleted directory	FTK Imager / Autopsy
Corel Draw Work	\corel draw work	Deleted directory	FTK Imager / Autopsy
SSH	\SSH	Deleted directory	FTK Imager / Autopsy
Buttons.cpt	...\Corel	Deleted image asset	Autopsy
DESIGN28.cpt	...\Corel	Deleted image asset	Autopsy
MSI9481b.tmp	\MSI9481b.tmp	Deleted temporary file	FTK Imager
desktop.ini	\$RECYCLE.BIN\S-1-5-21...	Active artifact	FTK Imager

Recovery caution A recovered file demonstrates that data remained recoverable from the image. It does not by itself establish who deleted the file, why it was deleted, or whether another copy existed before deletion.

14. Timeline Analysis

The supplied evidence supports a sequence rather than a single indisputable deletion timestamp. The earliest strong deletion-related marker in the record is the NTFS change timestamp of 16 September 2026 at 11:01:30 WAT. A later Recycle Bin artifact was recorded on 24 September 2026 at 11:42:43 WAT, followed by an access time of 11:45:41 WAT. Deleted-record access timestamps are also reported at 12:40:08 WAT.

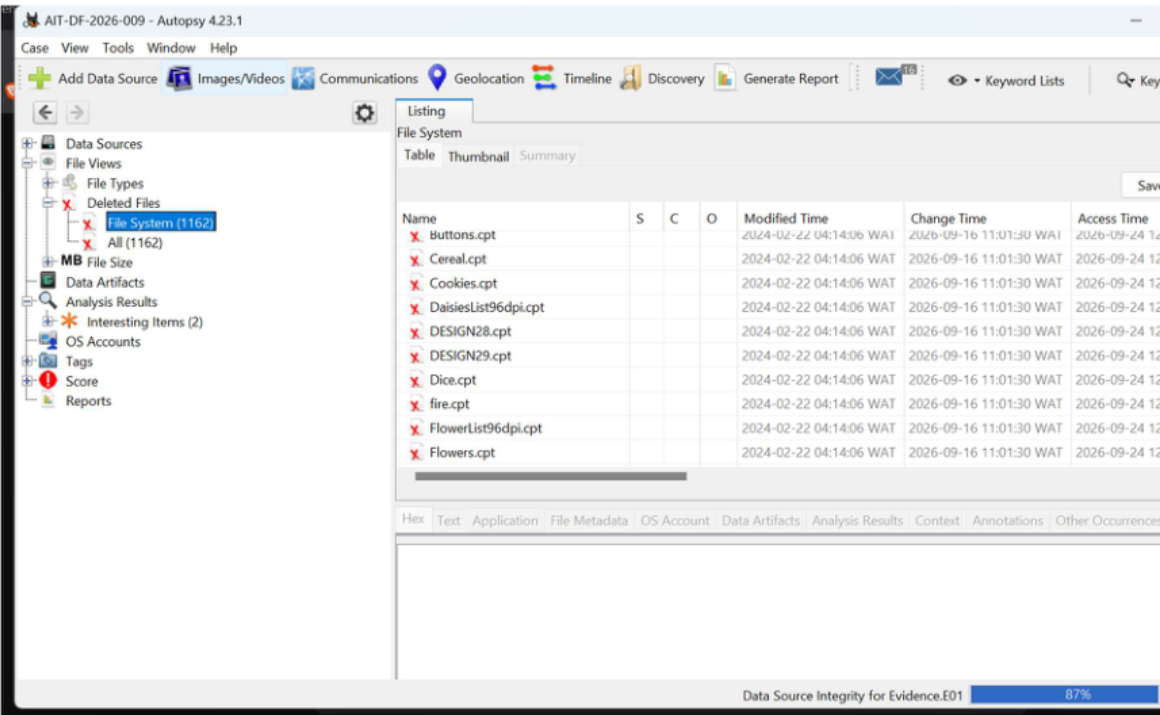


Figure 14.1 — Autopsy deleted-file view showing 1,162 results and associated timestamps. Reference exhibit from the supplied examination record.

Date / time (WAT)	Event represented	Evidence source	Interpretive weight
22 Feb 2024 04:14:06	Earlier creation/modification timestamps on deleted records	NTFS metadata	Shows prior filesystem history; not a deletion timestamp
16 Sep 2026 11:01:30	Change timestamp on relevant deleted records	NTFS \$MFT / Autopsy	Strong deletion-related boundary; not proof of keyboard action
24 Sep 2026 11:42:43	desktop.ini creation/modification artifact	\$RECYCLE.BIN	Later deletion-management / Recycle Bin-related activity
24 Sep 2026 11:45:41	desktop.ini access	\$RECYCLE.BIN	Subsequent access to the artifact
24 Sep 2026 12:40:08	Access timestamp across deleted records	Autopsy filesystem parser	Later metadata activity; not necessarily a deletion event

Defensible timeframe The supplied examination supports a deletion-related activity window from 16 September 2026 at 11:01:30 WAT through the later Recycle Bin artifact on 24 September 2026 at 11:42:43 WAT. The evidence does not justify reducing that entire window to a single proven human action.

15. Cross-Tool Validation

Finding	dd / R-Drive	FTK Imager	Autopsy	Interpretation
Image integrity	Hash / image handling recorded	E01 container examined	Data source integrity displayed	Image integrity was reported as verified
SSH/Corel/Corel Draw deletion	Underlying image sectors available	Deleted root MFT entries visible	Deleted records parsed	Multiple tools support deletion status
My pic / Our pic / ADDS	Raw image available	Unallocated areas visible	Carving / deleted-file parsing	Recovery depended on unallocated-space analysis
Recycle Bin state	Partition structure verified	SID folder and desktop.ini observed	Correlated in analysis	No target \$I/\$R records were reported
Graphic recovery	Raw data retained in image	MFT-linked export possible	Deleted .cpt assets parsed	Several graphic assets were recoverable

Differences between tools are expected because each tool exposes different layers of the same storage image. FTK Imager is particularly useful for direct filesystem and sector inspection, while Autopsy provides higher-level parsing, categorization and timeline views. A missing result in one interface should therefore not automatically be interpreted as evidence that the underlying data did not exist.

16. Forensic Questions — Answered

What evidence was supplied? Evidence.E01, an NTFS forensic disk image.

What is the evidence hash? SHA-256:

6216149FDDF6EB7074349AFAEF226F02C3B246EE6C3C3AE91C5AB7ADEB8C29E0; MD5:
ae39e9a30ab5a65809eb93289938e149.

Which target folders were deleted? All six are reported as deleted, but the filesystem presentation differs: SSH/Corel/Corel Draw Work retained deleted MFT directory records, while My pic/Our pic/ADDS Project were reported as deleted and unallocated.

Which deleted files were recovered? The supplied record reports recovery of Corel graphic assets and other deleted material, including Buttons.cpt and DESIGN28.cpt.

Why might some deleted material be unrecoverable? Overwritten sectors, fragmented data, missing filesystem metadata, or insufficient residual data can prevent complete reconstruction.

What evidence establishes the activity timeframe? NTFS change timestamps, deleted-record metadata, and the later \$RECYCLE.BIN desktop.ini artifact.

Can the physical user be identified? Not from the supplied disk evidence alone. A SID can associate artifacts with an account, but it does not establish who was physically operating the machine.

Can intentional deletion be established? The evidence establishes deletion-related filesystem changes; subjective intent is not established by the supplied artifacts alone.

Can copying before deletion be established? Not from the summarized evidence alone. Additional endpoint, shell, USB, cloud, or file-access artifacts would be needed.

What additional evidence would help? Windows Security/Event logs, endpoint telemetry, domain audit records, USB history, cloud synchronization logs, backups, and other relevant workstation artifacts.

17. Limitations

- The supplied record does not provide complete physical-workstation context.
- My pic, Our pic and ADDS Project were reported as unallocated, limiting direct reconstruction of original directory paths.
- The absence of \$I/\$R Recycle Bin records cannot, by itself, distinguish Shift+Delete from a later Recycle Bin purge.
- Filesystem timestamps describe metadata states; they do not automatically identify the human action that produced them.
- The summarized record does not independently demonstrate whether files were copied elsewhere before deletion.
- Some reported tool activity and recovered-file hashes should be reproduced or verified before being represented as the investigator's own processing results.

18. Findings

- Evidence.E01 was reported as an NTFS forensic image whose SHA-256 and MD5 values matched the recorded working-copy values.
- All six target folders were reported as deleted, with different levels of surviving filesystem structure.
- SSH, Corel and Corel Draw Work retained deleted root-level MFT entries in the supplied FTK examination.
- My pic, Our pic and ADDS Project were reported as deleted/unallocated and required carving or related artifact analysis.
- Autopsy reported 1,162 deleted filesystem entries.
- Multiple Corel .cpt assets were reported as recoverable.
- The strongest deletion-related timestamp was 16 September 2026 at 11:01:30 WAT, with later Recycle Bin-related activity on 24 September 2026.

19. Conclusion

The available forensic record supports the conclusion that the six target folders previously existed on the examined NTFS volume and were subsequently removed from their active filesystem locations. Surviving MFT records permitted direct identification of three deleted root-level directories, while the other three were reported in unallocated space and required recovery-oriented analysis.

The evidence further shows that a substantial amount of deleted material remained recoverable. The timeline is best expressed as a bounded period of deletion-related activity beginning no later than the 16 September 2026 MFT change timestamp and extending to the later Recycle Bin artifact on 24 September 2026. The supplied disk evidence does not establish the physical person responsible or their subjective intent.

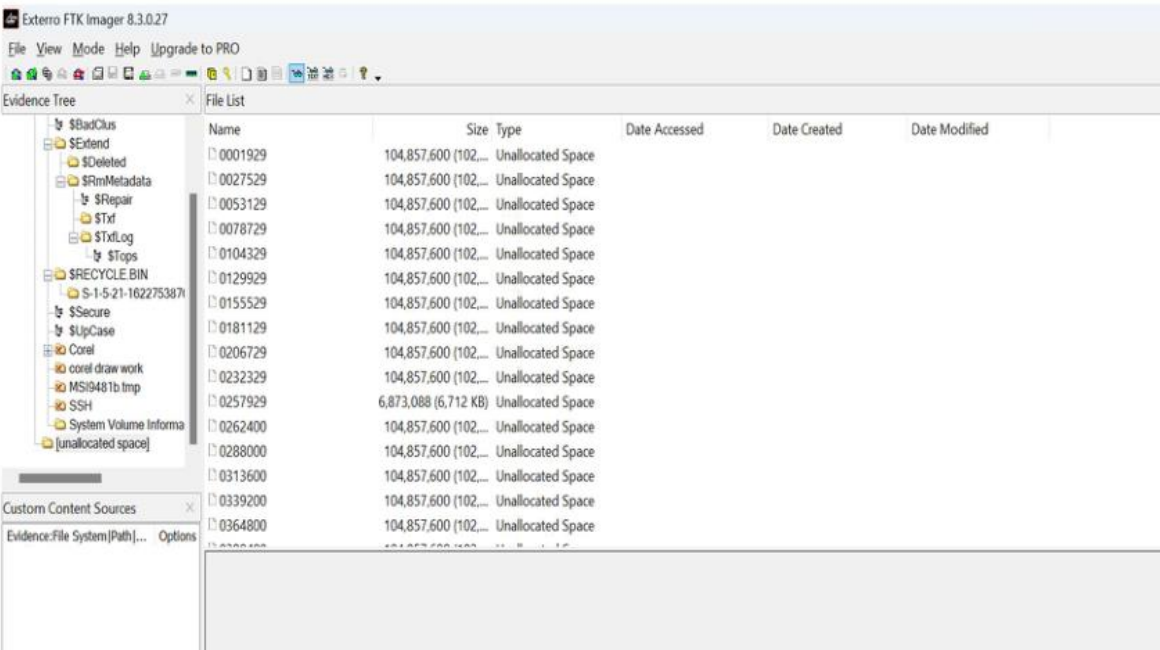
20. Recommendations

- Enable Windows object-access auditing for sensitive directories, including relevant Event IDs 4660 and 4663 where appropriate.
- Centralize endpoint and security logs in tamper-resistant storage.
- Maintain documented evidence-handling and hashing procedures for future investigations.
- Preserve endpoint telemetry, backup records, USB history and domain audit logs when investigating suspected deletion.

Appendix A — Evidence Exhibits

The following screenshots are preserved as reference exhibits from the supplied examination record. If you perform the examination yourself, replace these with your own captures and retain the same analytical captions where applicable.

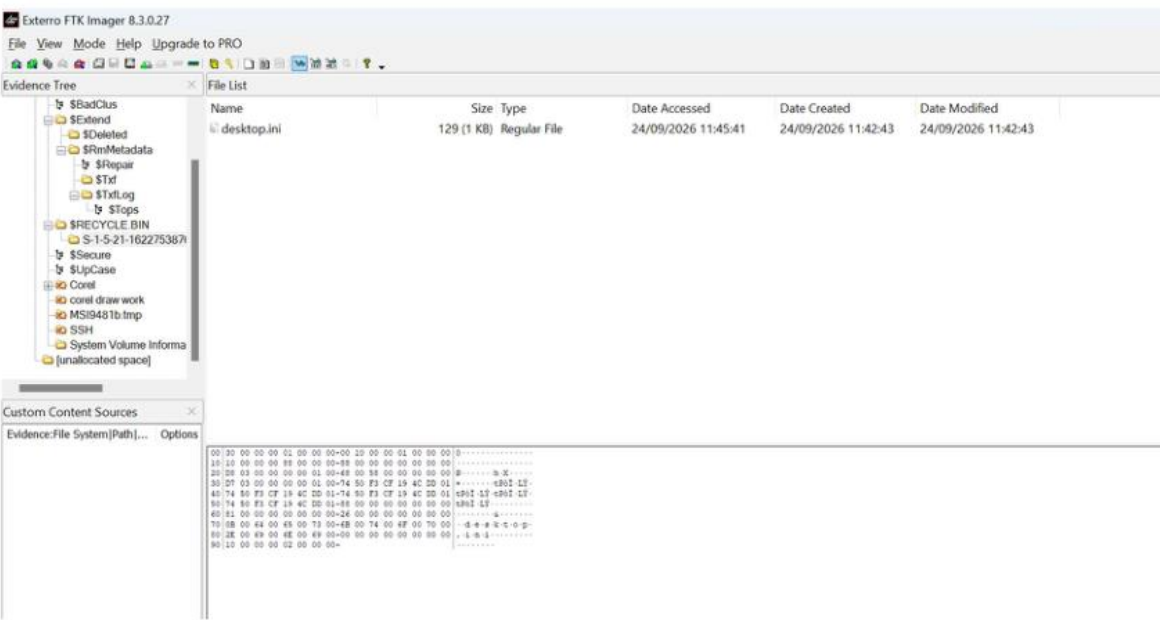
Exhibit A1 — Unallocated-space view



FTK Imager view showing unallocated-space entries and the filesystem tree.

Verification status: replace/confirm against your own examination capture before final submission.

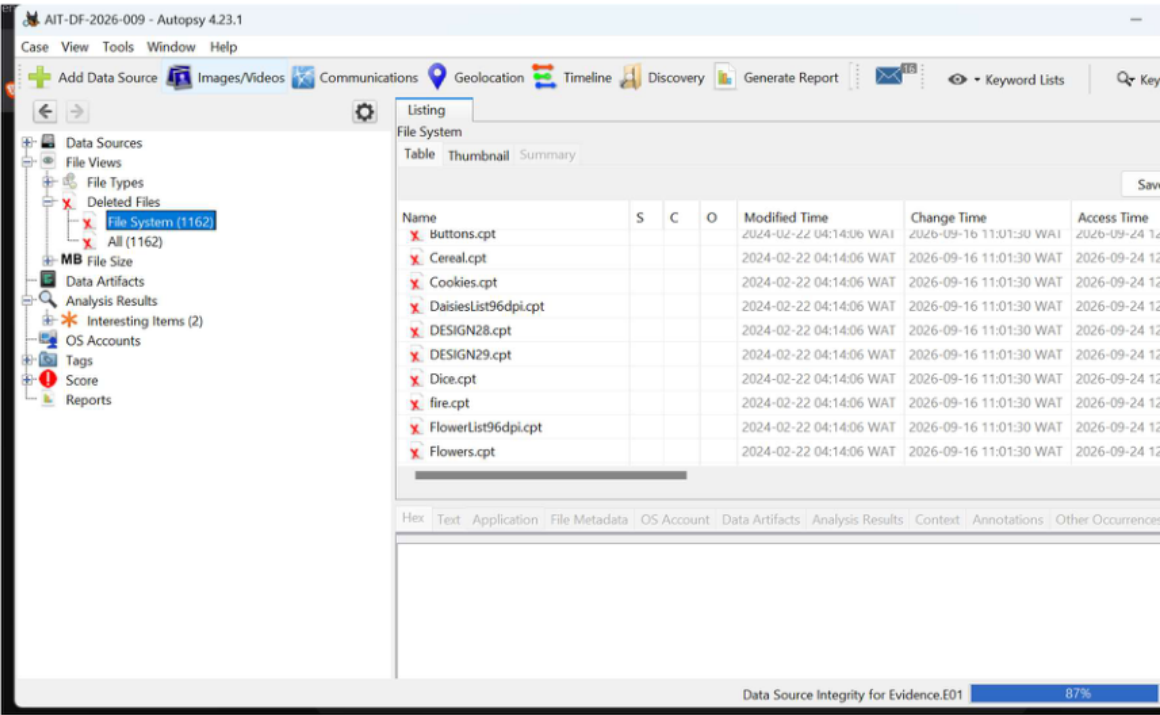
Exhibit A2 — Recycle Bin artifact



FTK Imager view showing the user SID directory with desktop.ini.

Verification status: replace/confirm against your own examination capture before final submission.

Exhibit A3 — Autopsy deleted-file view



Autopsy view showing deleted-file results and associated metadata timestamps.

Verification status: replace/confirm against your own examination capture before final submission.

Appendix B — Investigation Log

Date / time	Tool	Action	Recorded result
24 Sep 2026 09:30	dd / Terminal	Calculated SHA-256 and MD5	Verification matched recorded source values
24 Sep 2026 10:15	FTK Imager 8.3	Mounted Evidence.E01 and inspected root	Deleted SSH, Corel and corel draw work identified
24 Sep 2026 11:00	FTK Imager 8.3	Inspected \$RECYCLE.BIN	desktop.ini artifact identified
24 Sep 2026 11:30	FTK Imager 8.3	Inspected unallocated space	Raw sector chunks catalogued
24 Sep 2026 12:40	Autopsy 4.23.1	Processed filesystem and deleted records	1,162 deleted entries reported
27 Sep 2026 20:00	Autopsy / CLI	Exported files and calculated output hashes	hashes.txt reported

Appendix C — Evidence Processing Table

Evidence	Original path	Status	Relevant time	Recovery	Tool
Corel	\Corel	Deleted	16 Sep 2026 11:01:30	Recoverable	FTK / Autopsy
Corel Draw Work	\corel draw work	Deleted	16 Sep 2026 11:01:30	Recoverable	FTK / Autopsy
SSH	\SSH	Deleted	16 Sep 2026 11:01:30	Recoverable	FTK / Autopsy
Buttons.cpt	...\Corel	Deleted	16 Sep 2026 11:01:30	Recovered	Autopsy
DESIGN28.cpt	...\Corel	Deleted	16 Sep 2026 11:01:30	Recovered	Autopsy
desktop.ini	\$RECYCLE.BIN\S-1-5-21...	Active artifact	24 Sep 2026 11:42:43	Recovered	FTK Imager

Appendix D — Submission Checklist

- Final PDF report
- Evidence/recovery folder
- Hash record for exported/recovered files
- Clearly labelled screenshots
- Timeline table and/or visual timeline
- Completed investigation log
- Completed chain-of-custody record
- Independent verification of every result presented as personally performed